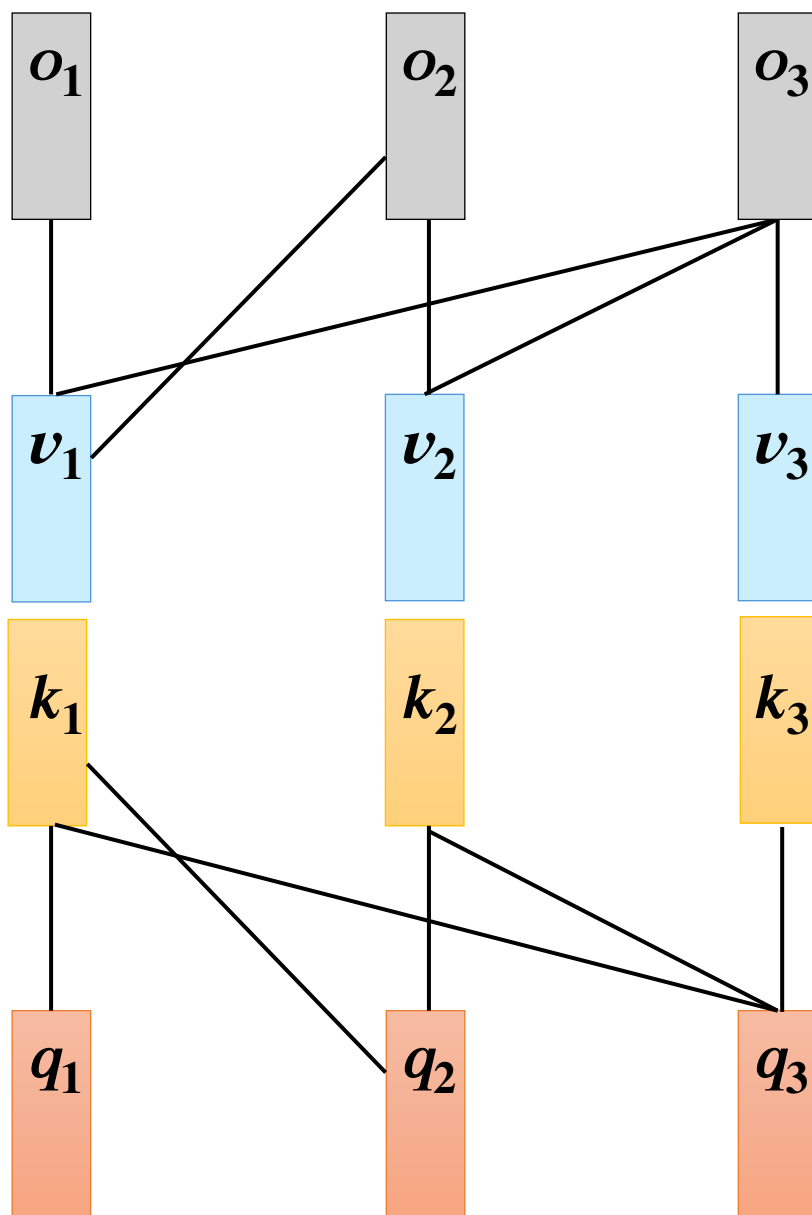
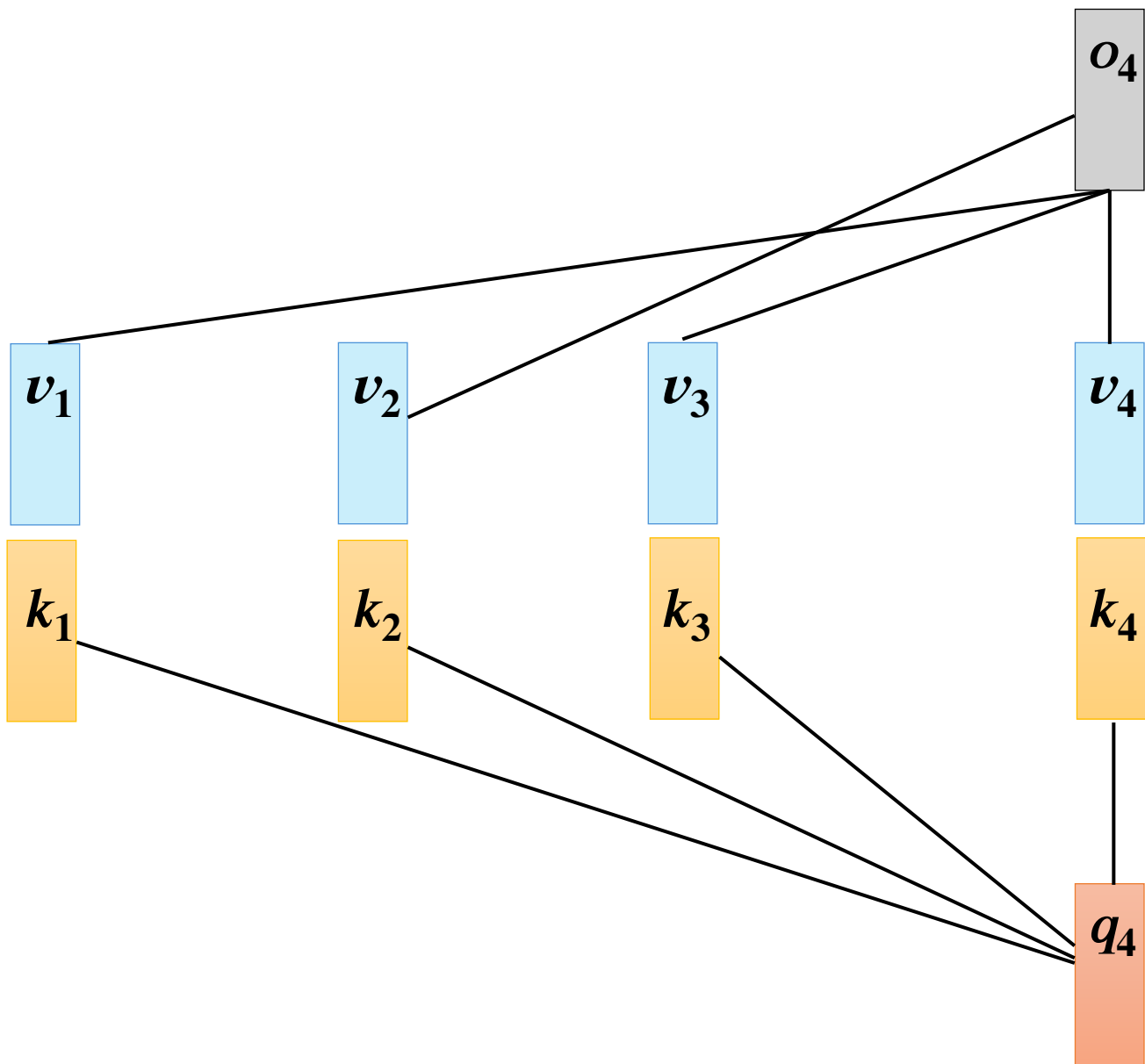
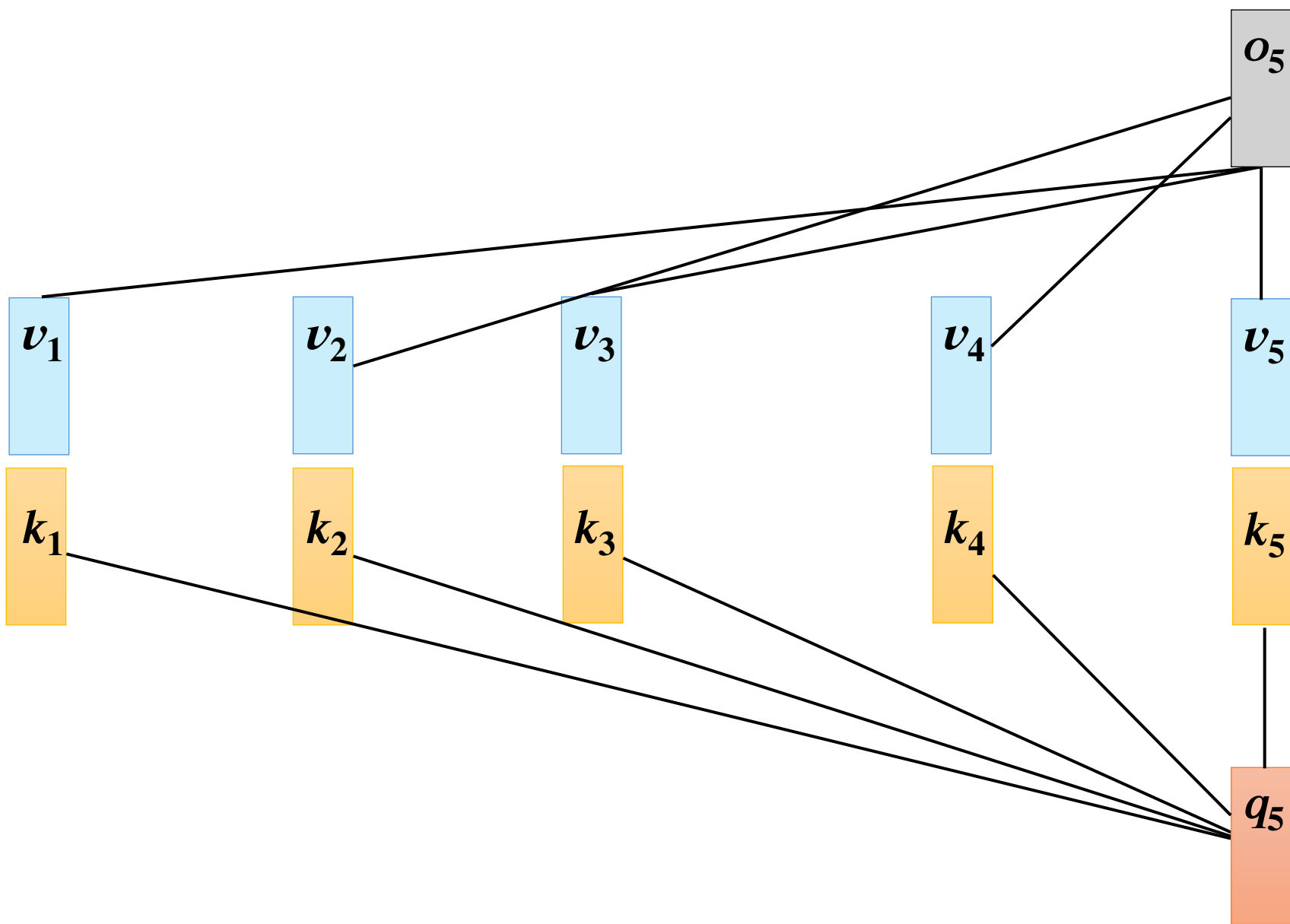


KV Cache









HBM



SRAM

Gemma 2

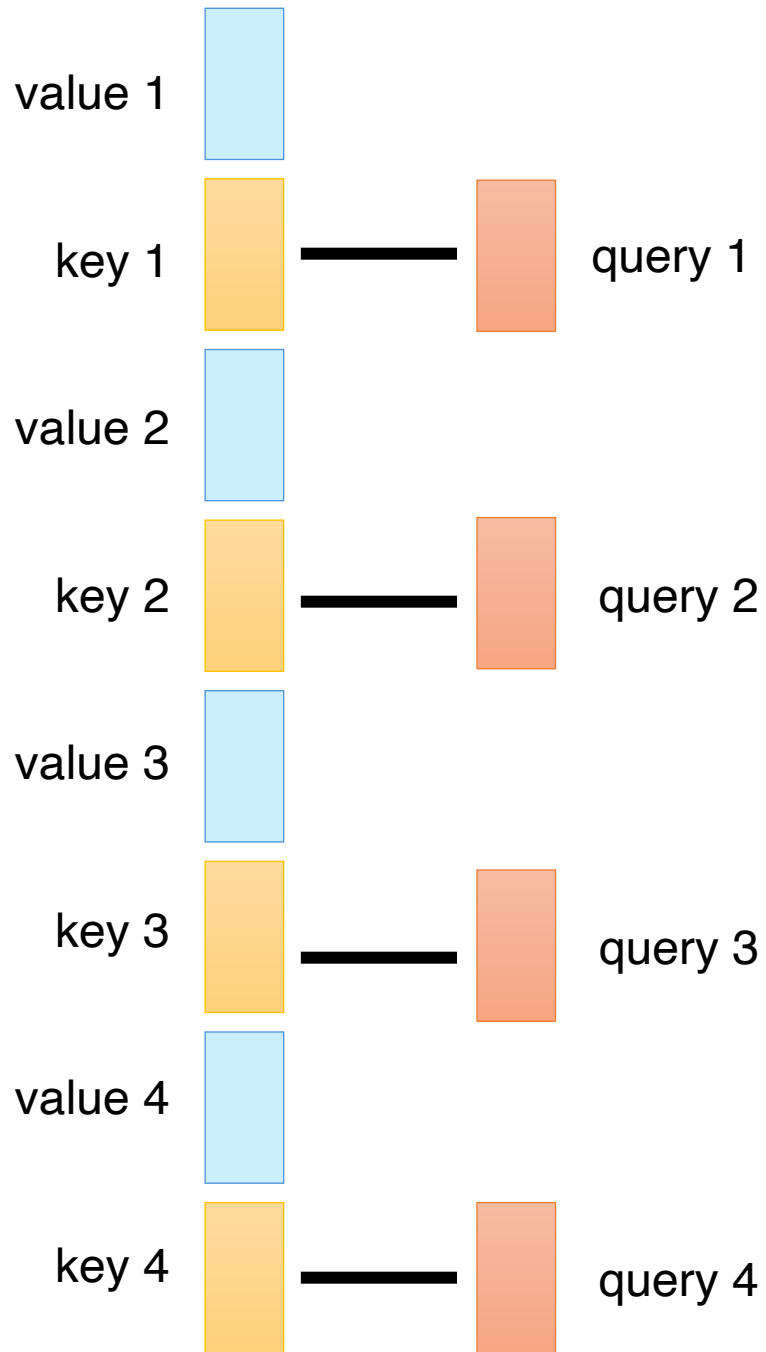
Parameters	2B	9B	27B
d_{model}	2304	3584	4608
Layers	26	42	46
Pre-norm	yes	yes	yes
Post-norm	yes	yes	yes
Non-linearity	GeGLU	GeGLU	GeGLU
Feedforward dim	18432	28672	73728
Head type	GQA	GQA	GQA
Num heads	8	16	32
Num KV heads	4	8	16
Head size	256	256	128
Global att. span	8192	8192	8192
Sliding window	4096	4096	4096
Vocab size	256128	256128	256128
Tied embedding	yes	yes	yes

Each token:

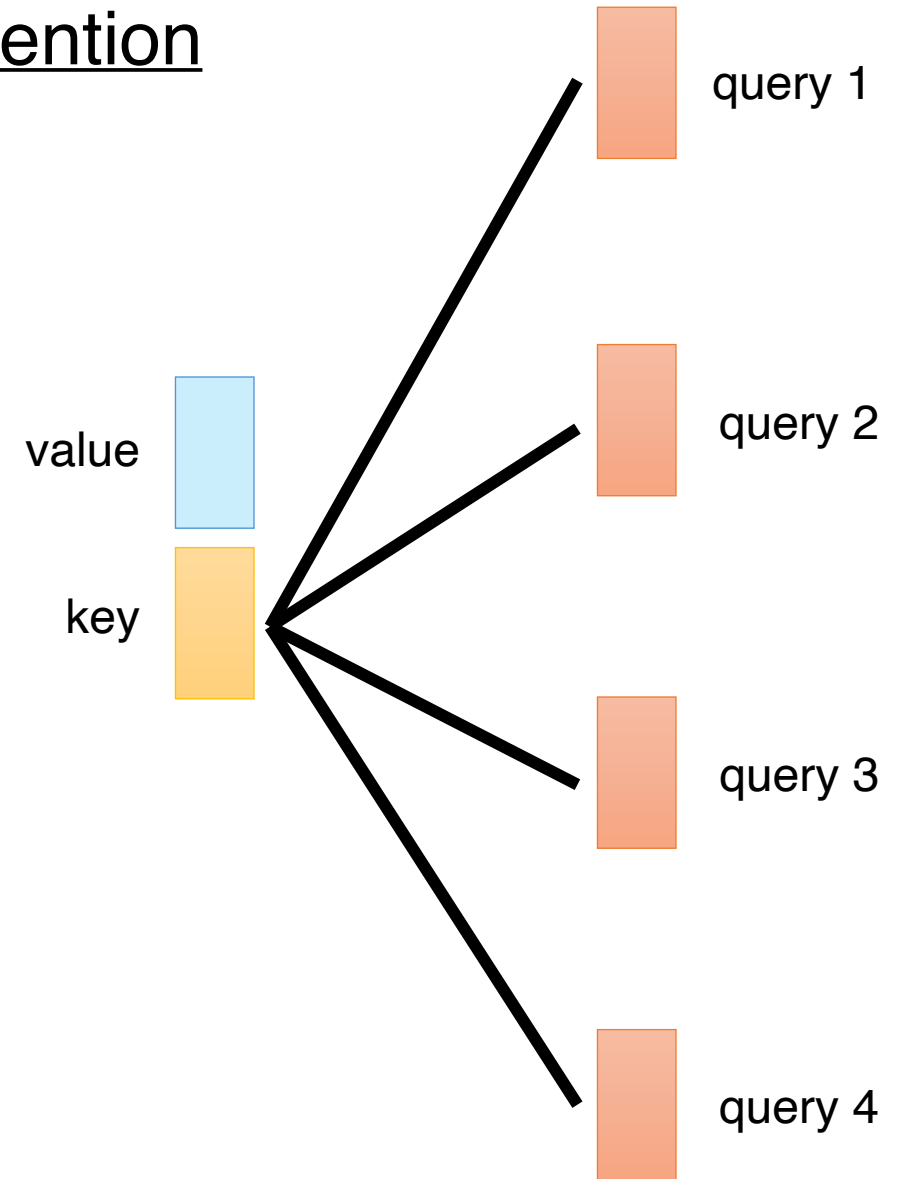
46 (layers) x 32 (heads) x 128
(dimension) x 2 (FP16) x 2 (value, key)
= 753664 bytes (736 KB, 0.72MB)

A100 (80GB):

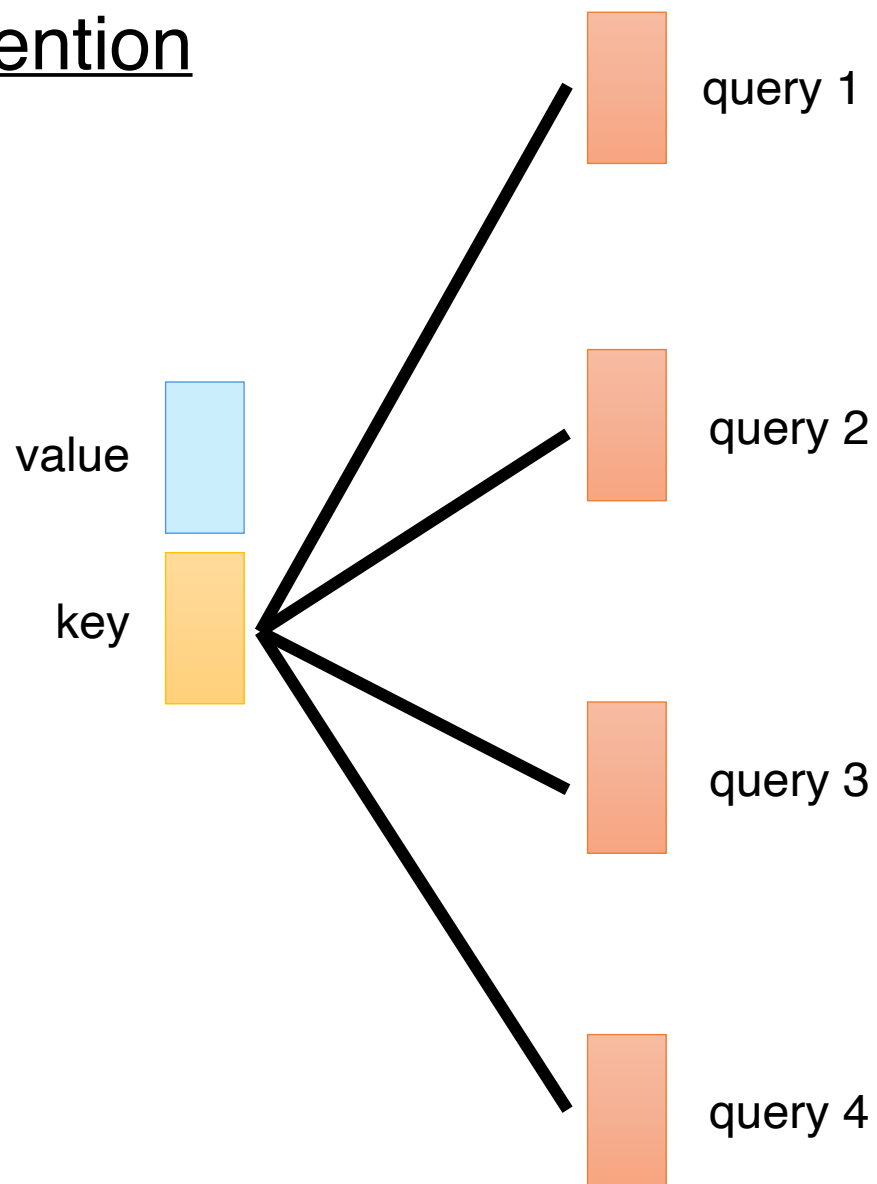
About 114k tokens



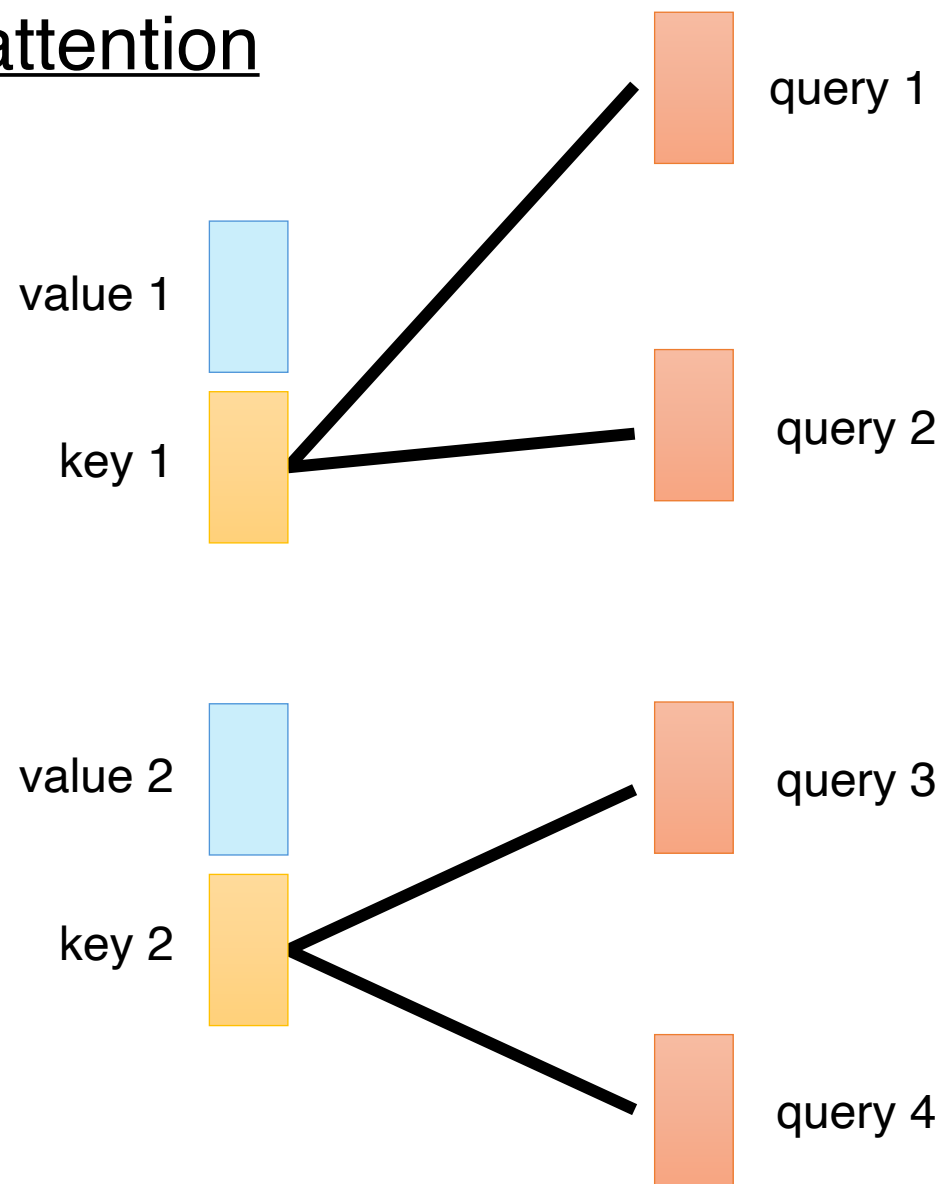
Multi-query attention



Multi-query attention



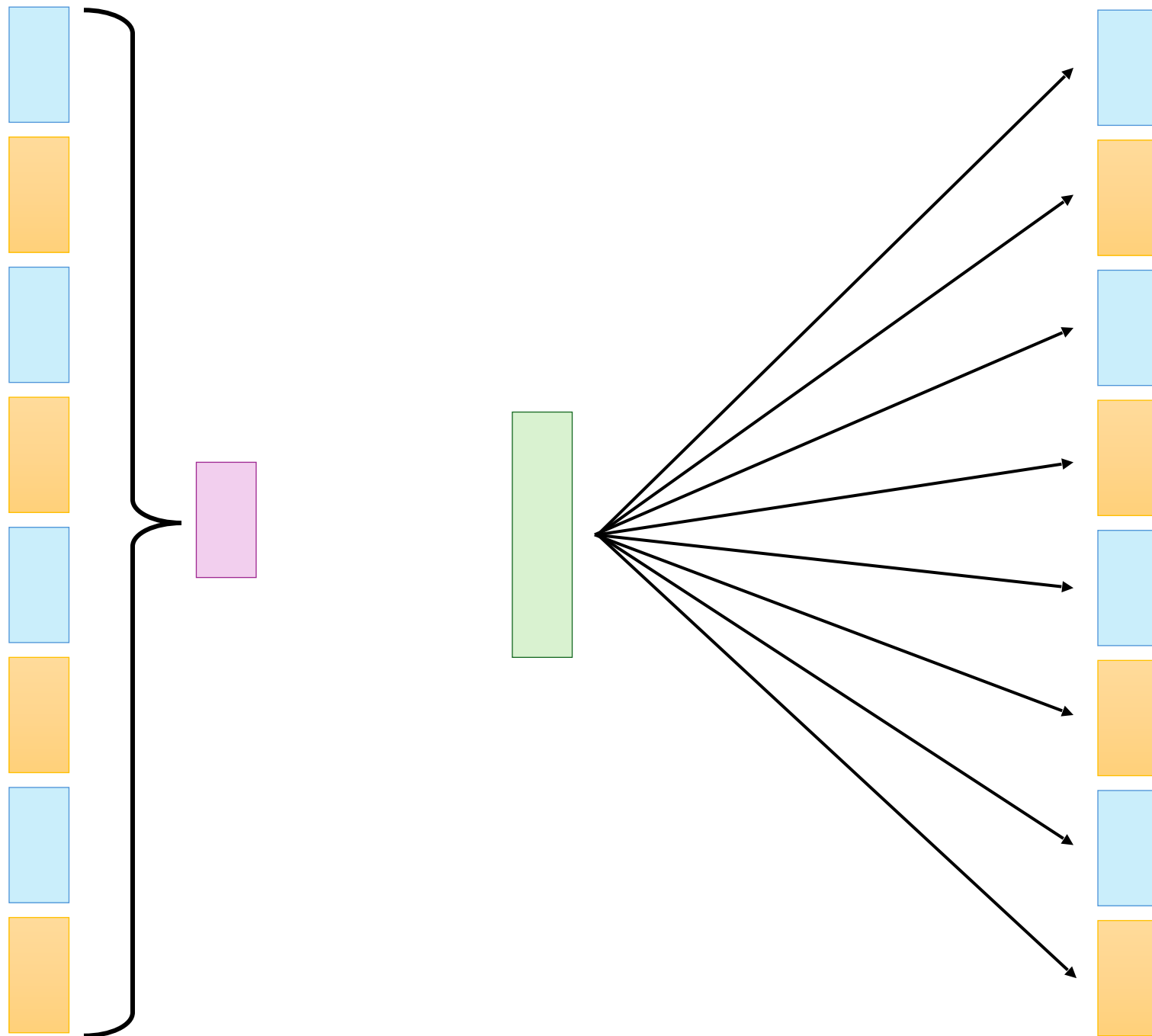
Group-query attention



Llama, Gemma, etc.

Multi-head Latent Attention

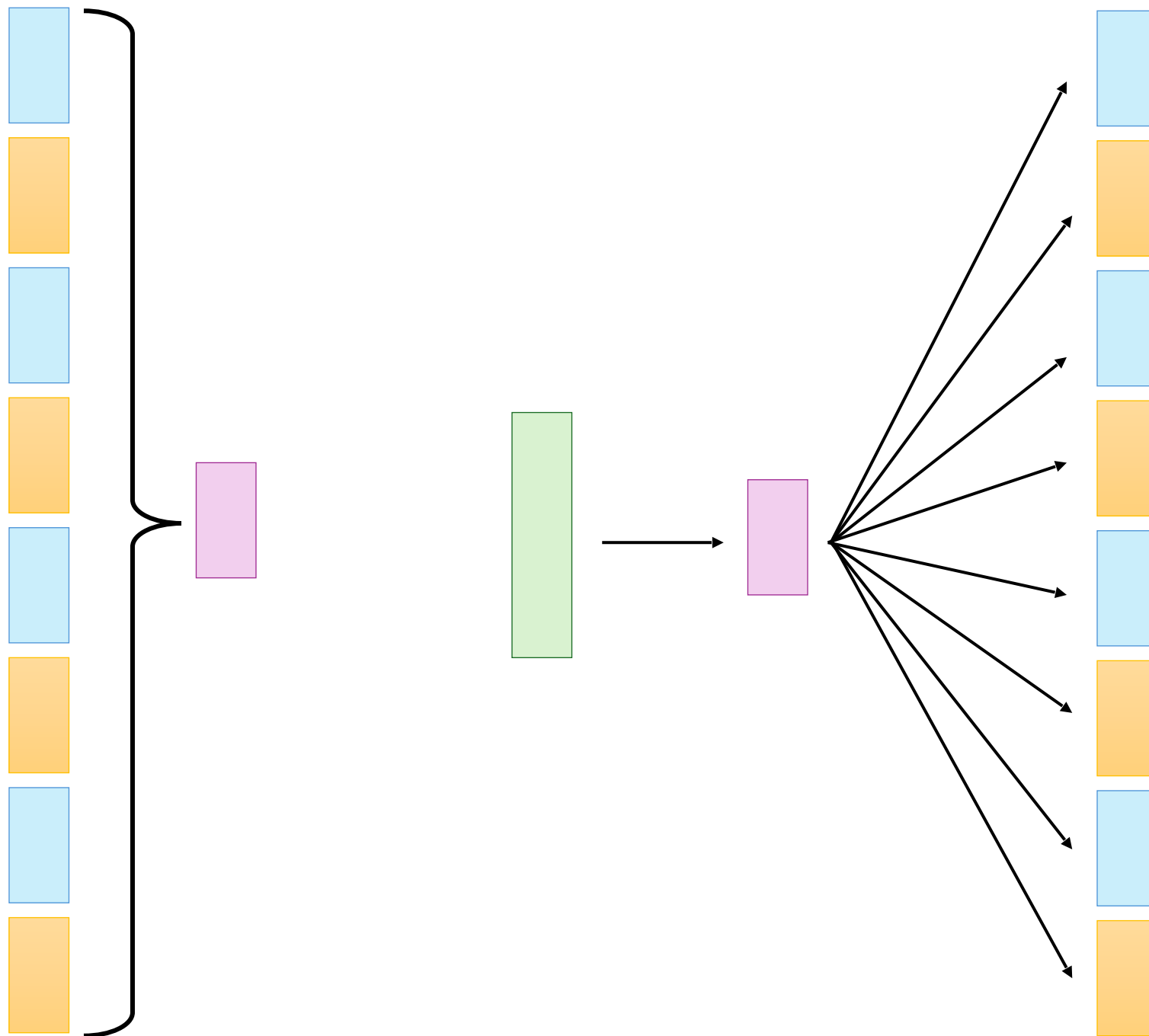
[https://arxiv.org/
abs/2405.04434](https://arxiv.org/abs/2405.04434)

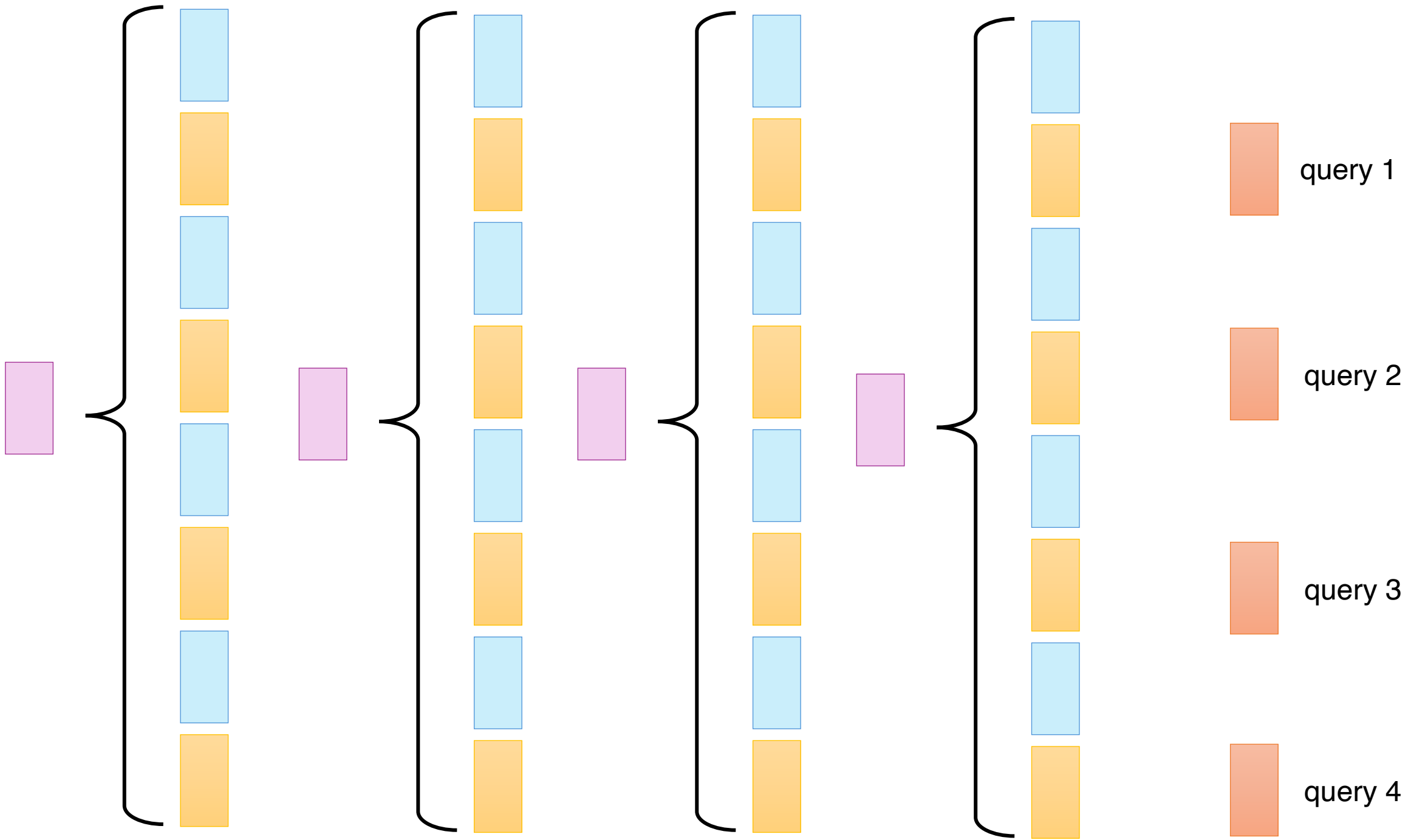


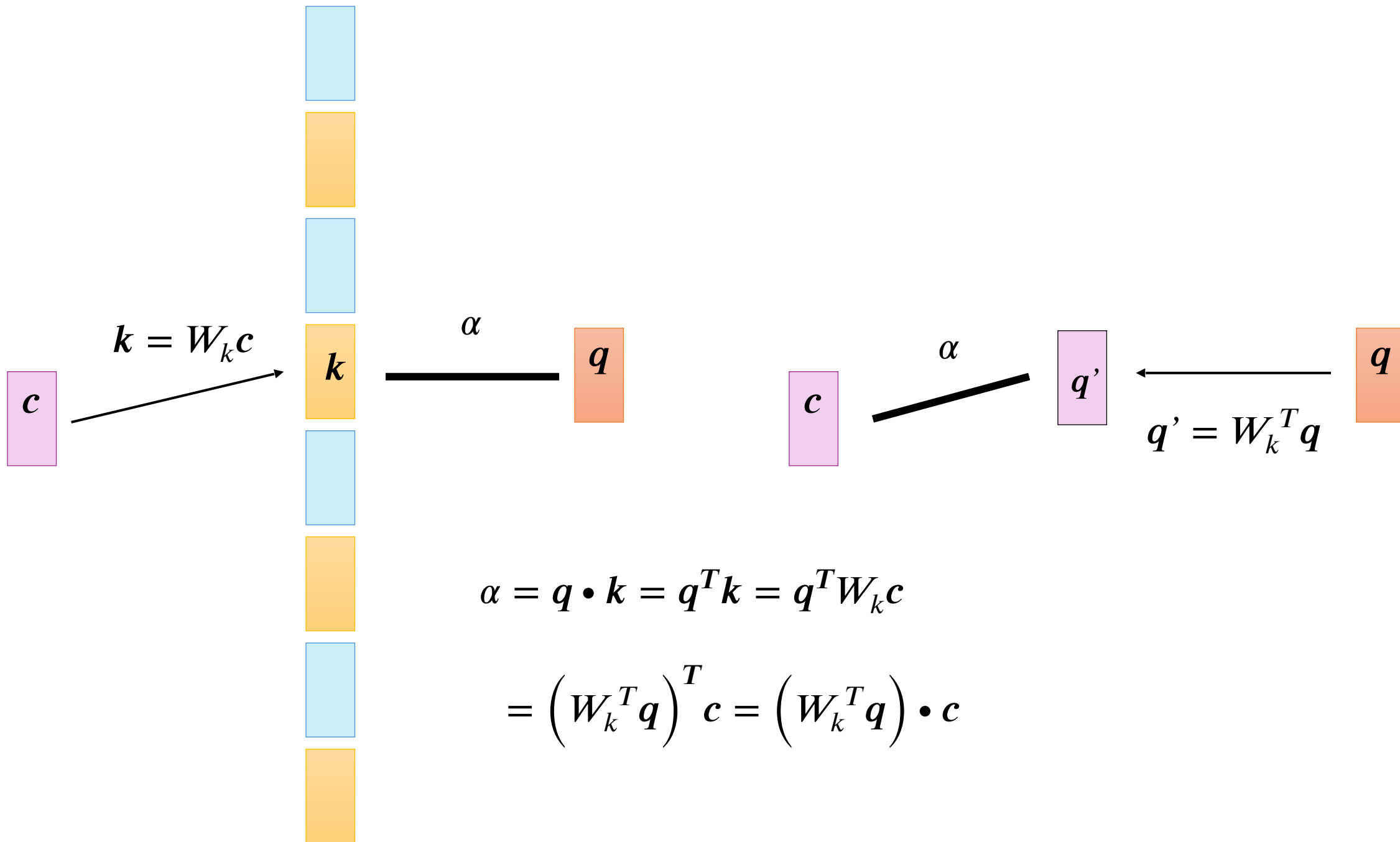
Multi-head Latent Attention

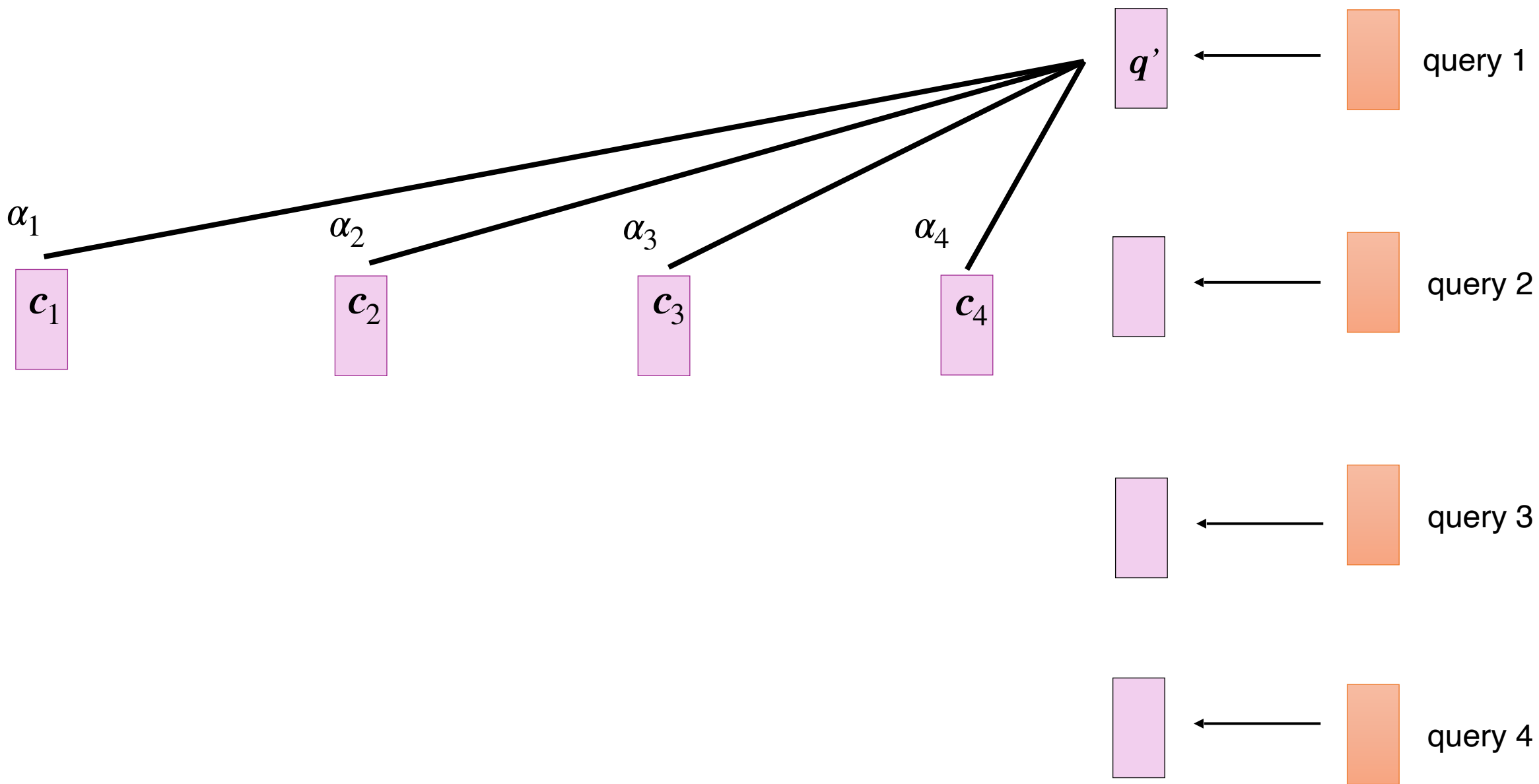
DeepSeek

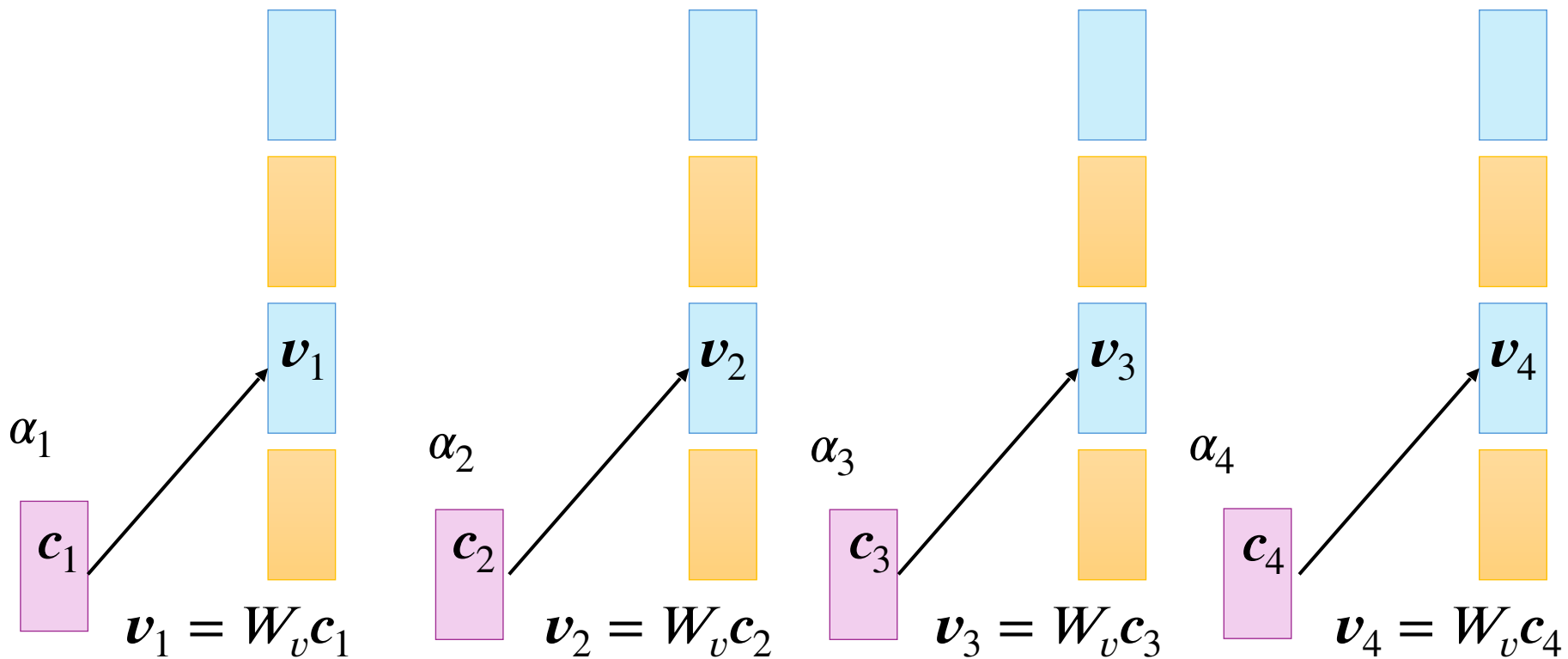
[https://arxiv.org/
abs/2405.04434](https://arxiv.org/abs/2405.04434)









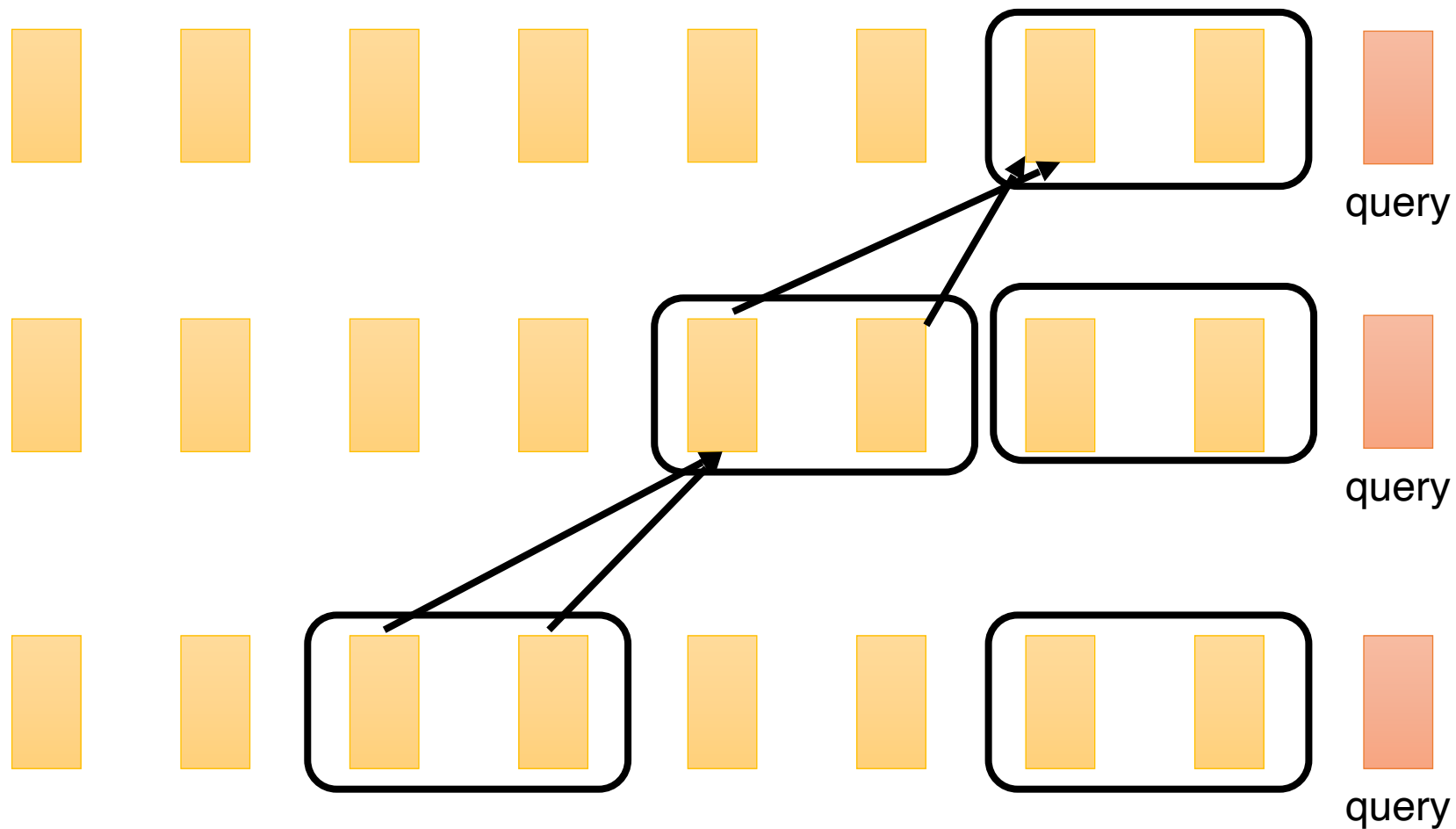


$$\mathbf{o} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 + \alpha_4 \mathbf{v}_4$$

$$= \alpha_1 W_v \mathbf{c}_1 + \alpha_2 W_v \mathbf{c}_2 + \alpha_3 W_v \mathbf{c}_3 + \alpha_4 W_v \mathbf{c}_4$$

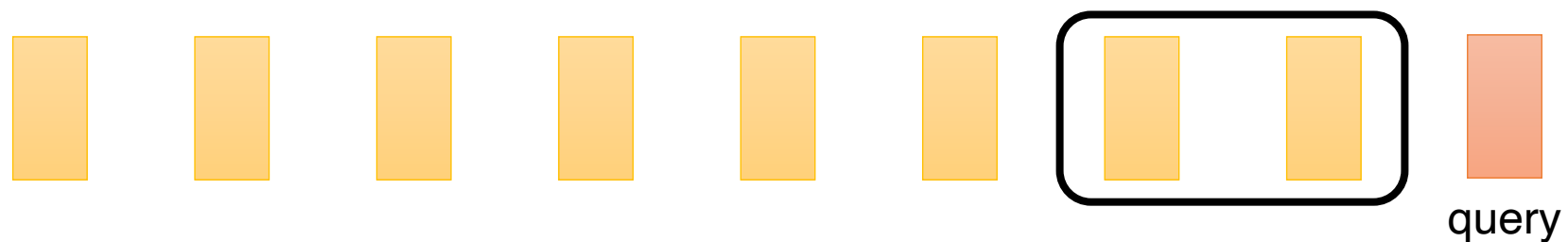
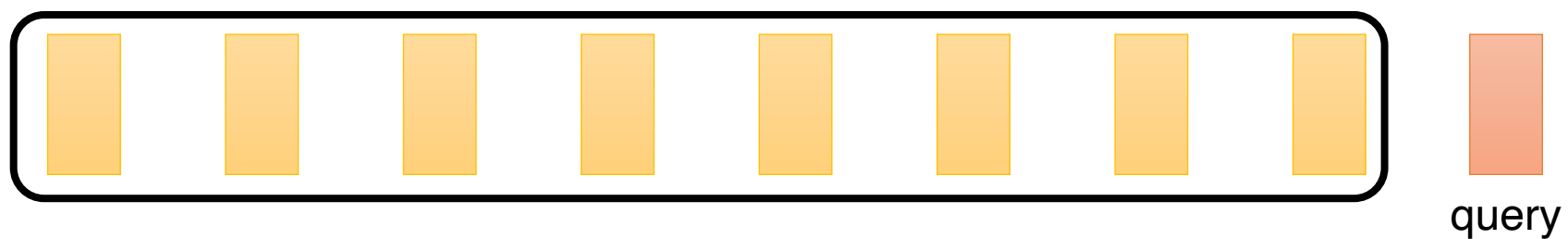
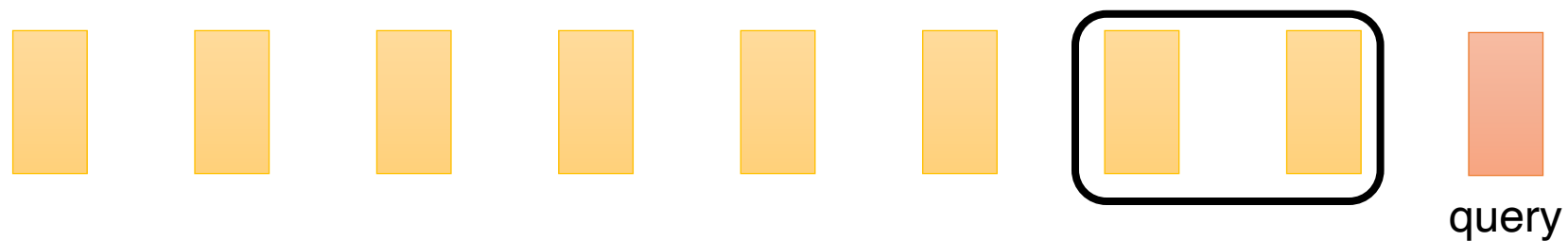
$$= W_v (\alpha_1 \mathbf{c}_1 + \alpha_2 \mathbf{c}_2 + \alpha_3 \mathbf{c}_3 + \alpha_4 \mathbf{c}_4)$$

Sliding Window Attention



Mistral 7B

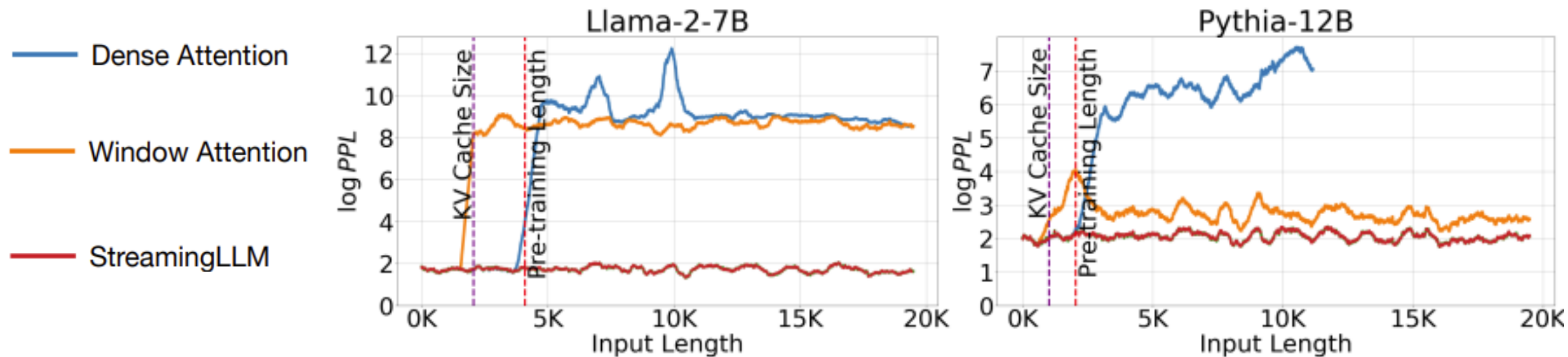
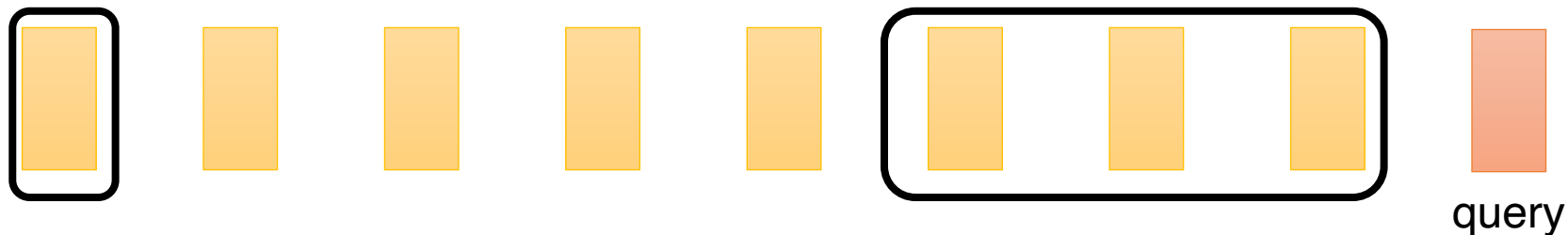
Sliding Window Attention



GPT-OSS

Streaming LLM

<https://arxiv.org/abs/2309.17453>

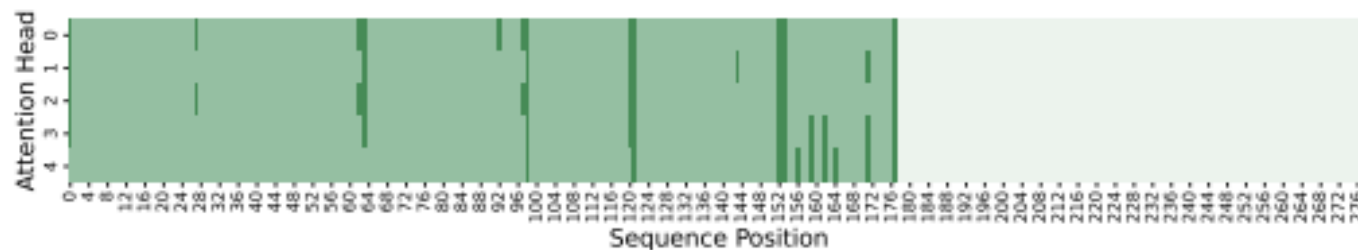


Pruning KV Cache

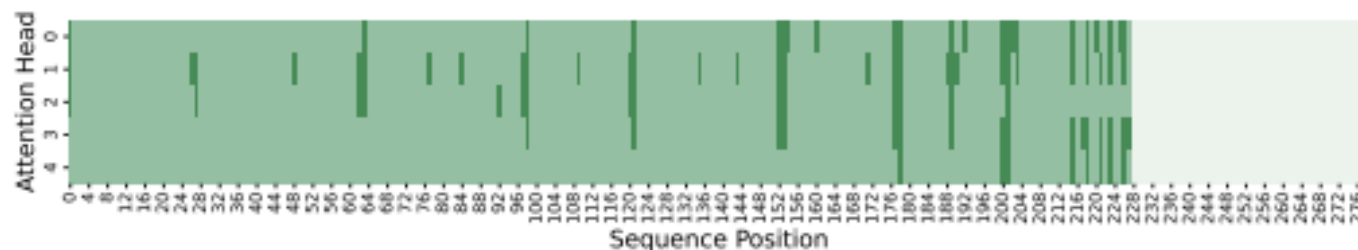
Scissorhands: <https://arxiv.org/abs/2305.17111>

H2O: <https://arxiv.org/abs/2306.14048>

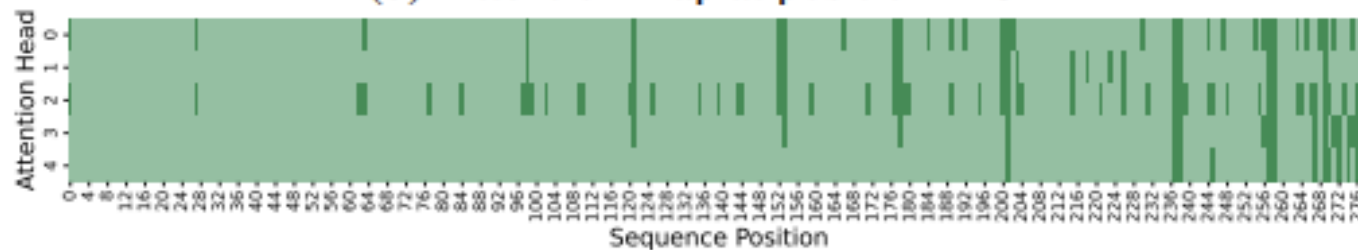
- 每次只有小部分的 Tokens 有 Attention
- 少數 token 會反覆吸走大量 attention



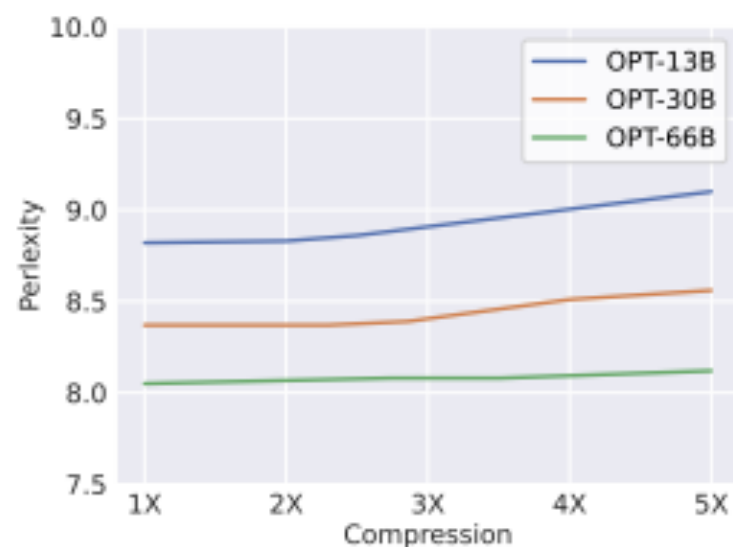
(a) Attention map at position 178



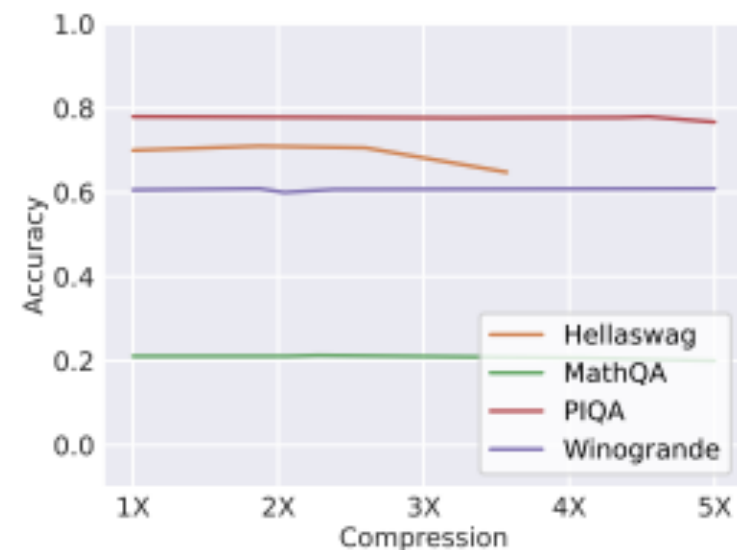
(b) Attention map at position 228



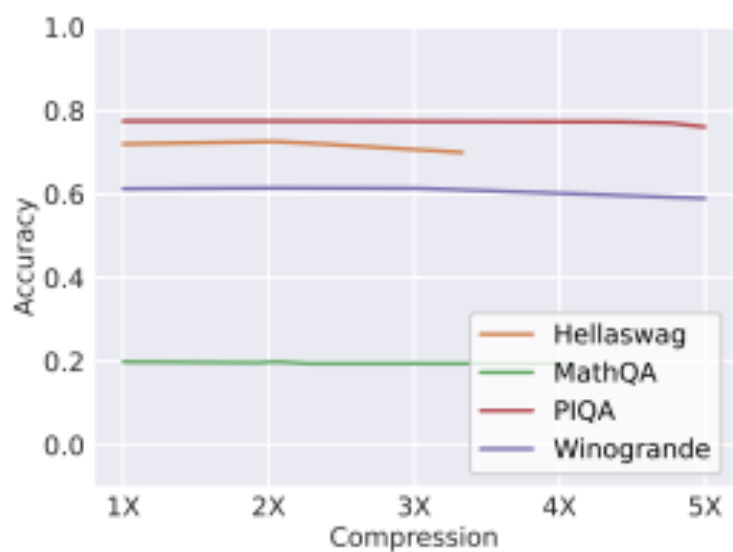
(c) Attention map at position 278



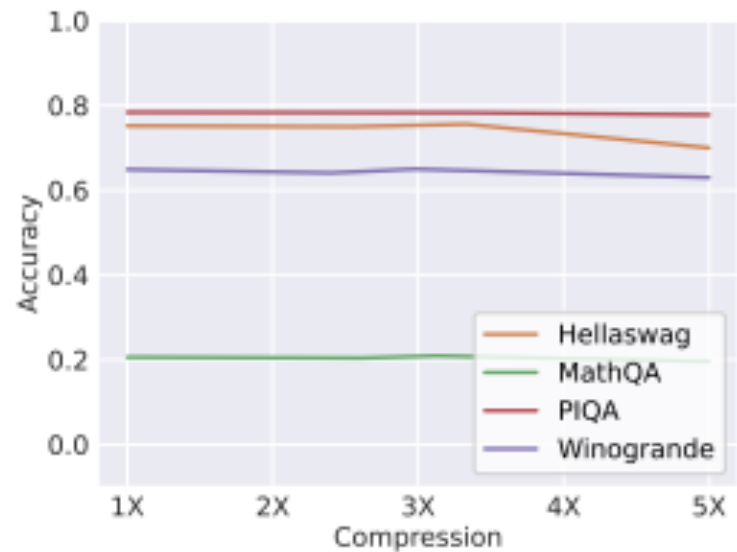
(a) Language Modeling



(b) OPT-6B Five shot



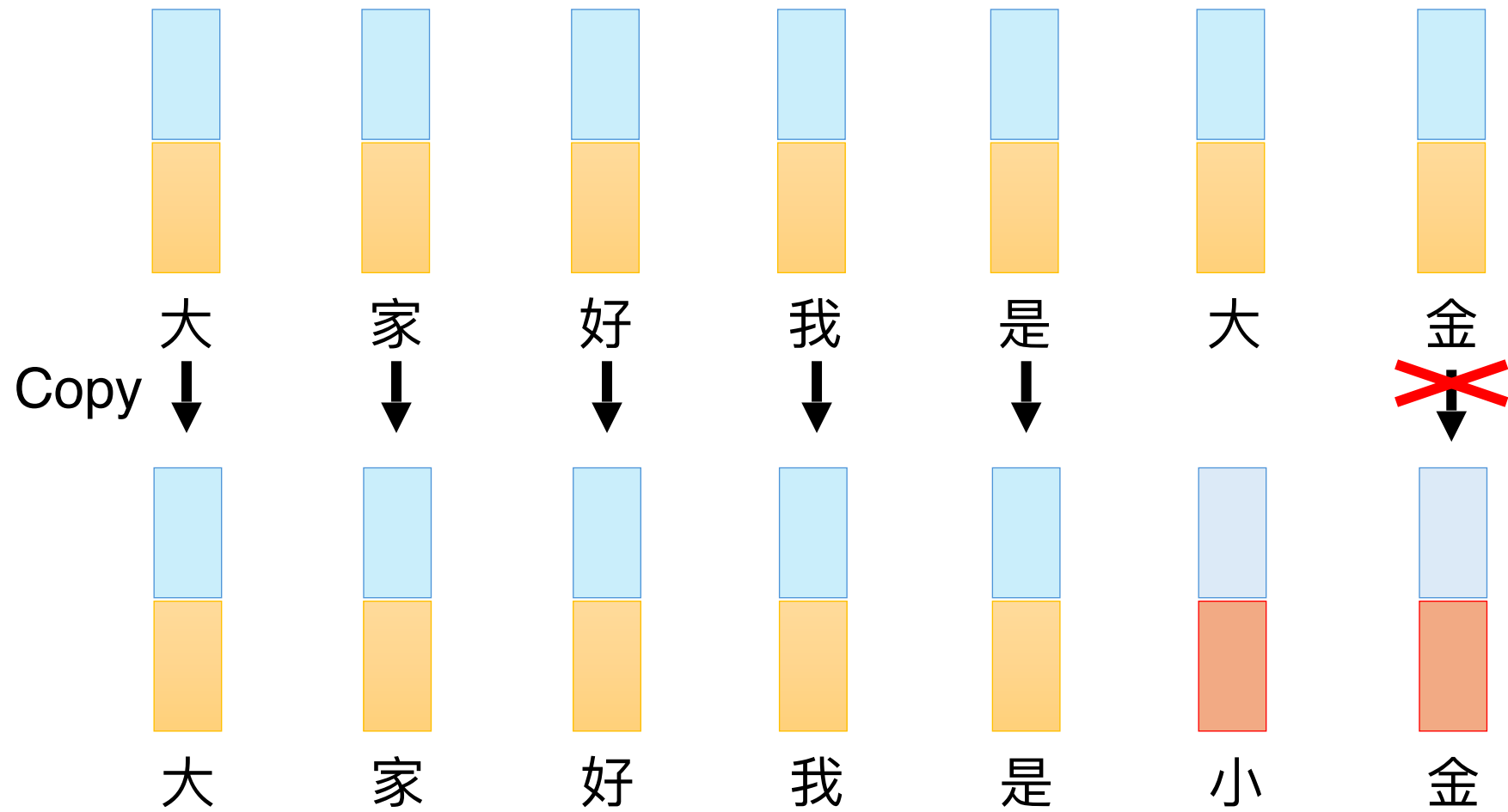
(c) OPT-13B Five shot



(d) OPT-30B Five shot

Scissorhands: <https://arxiv.org/abs/2305.17118>

跨對話的 Cache



Text tokens

Prices per 1M tokens.

Standard

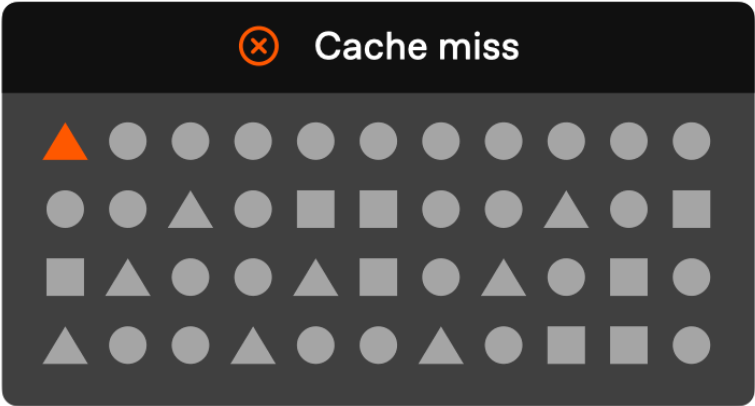
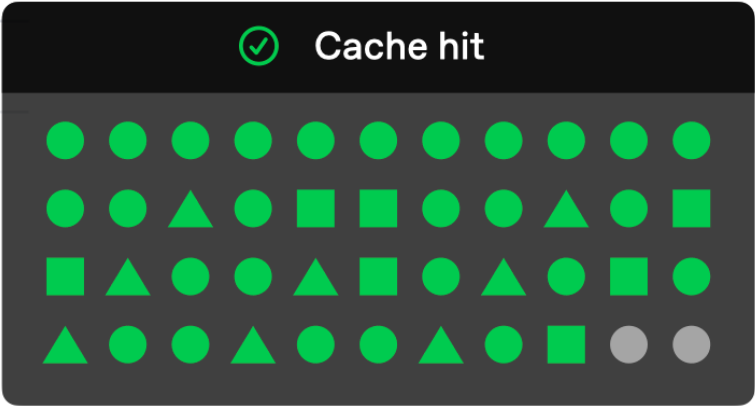
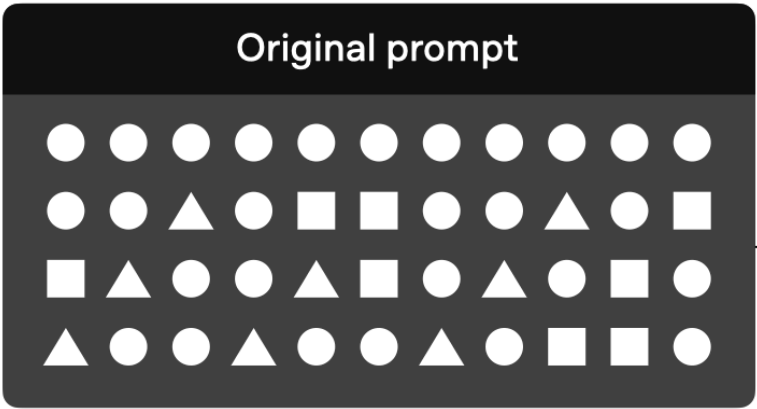
Batch

Flex

Priority

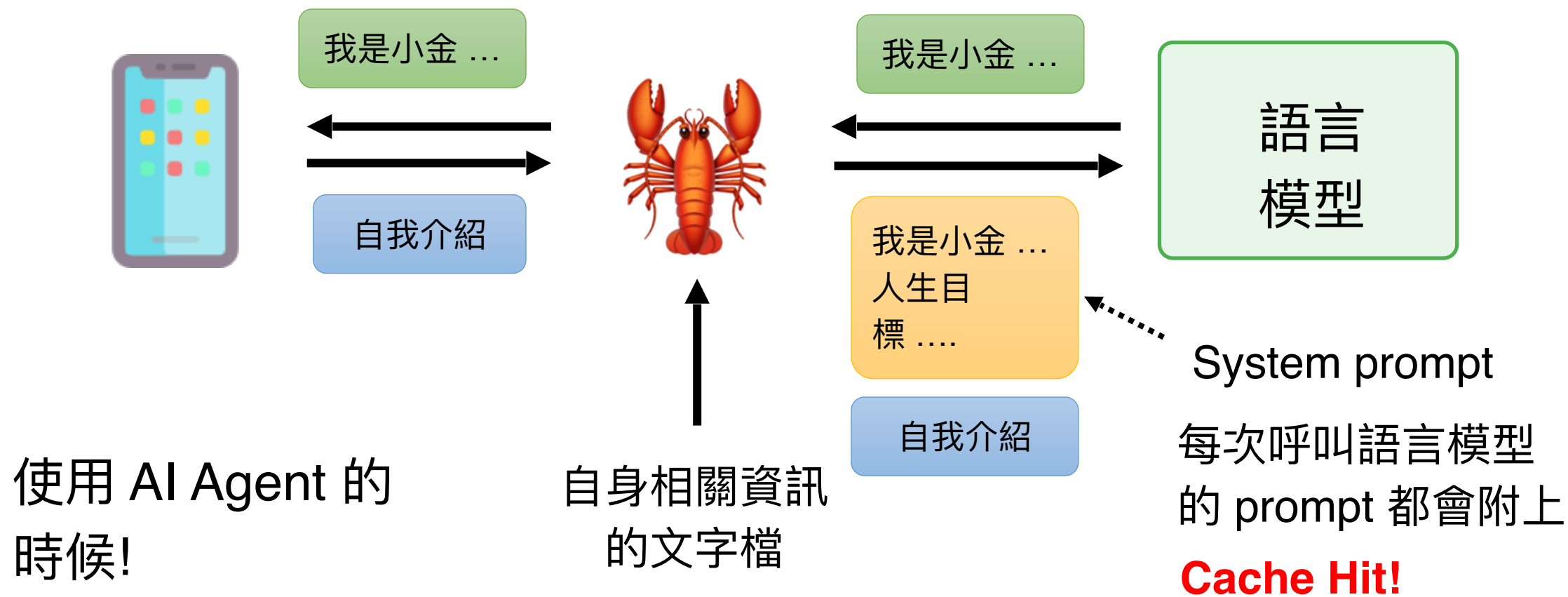
Our latest models

Model	Short context ⓘ			Long context ⓘ		
	Input	Cached input	Output	Input	Cached input	Output
gpt-5.4	\$2.50	\$0.25	\$15.00	\$5.00	\$0.50	\$22.50
gpt-5.4-pro	\$30.00	-	\$180.00	\$60.00	-	\$270.00

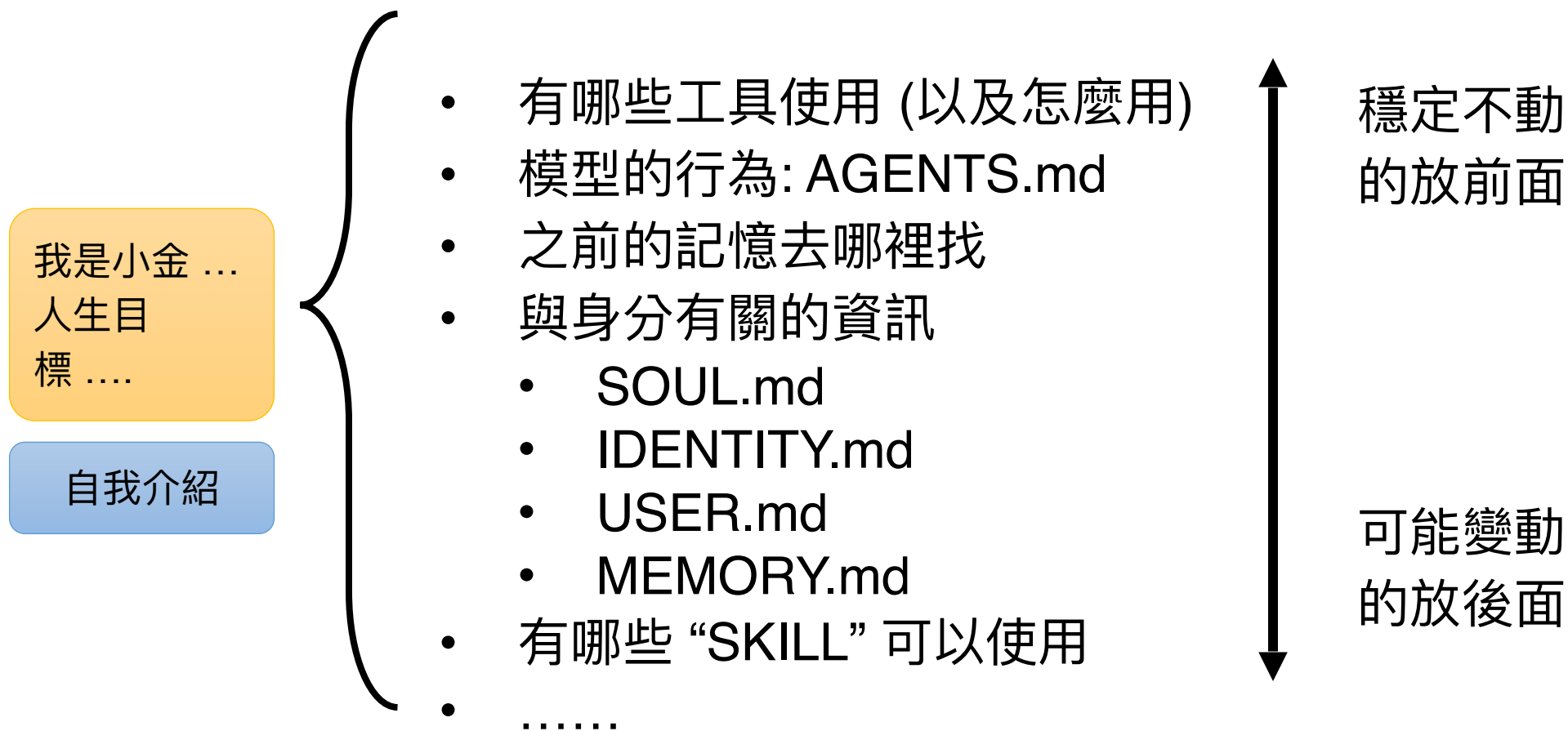


<https://developers.openai.com/api/docs/guides/prompt-caching/>
<https://developers.openai.com/api/docs/pricing>

甚麼時候跨對話的 Cache 可以發揮作用



甚麼時候跨對話的 Cache 可以發揮作用



Cache hit little



幫我訂從台北到波士頓的班機

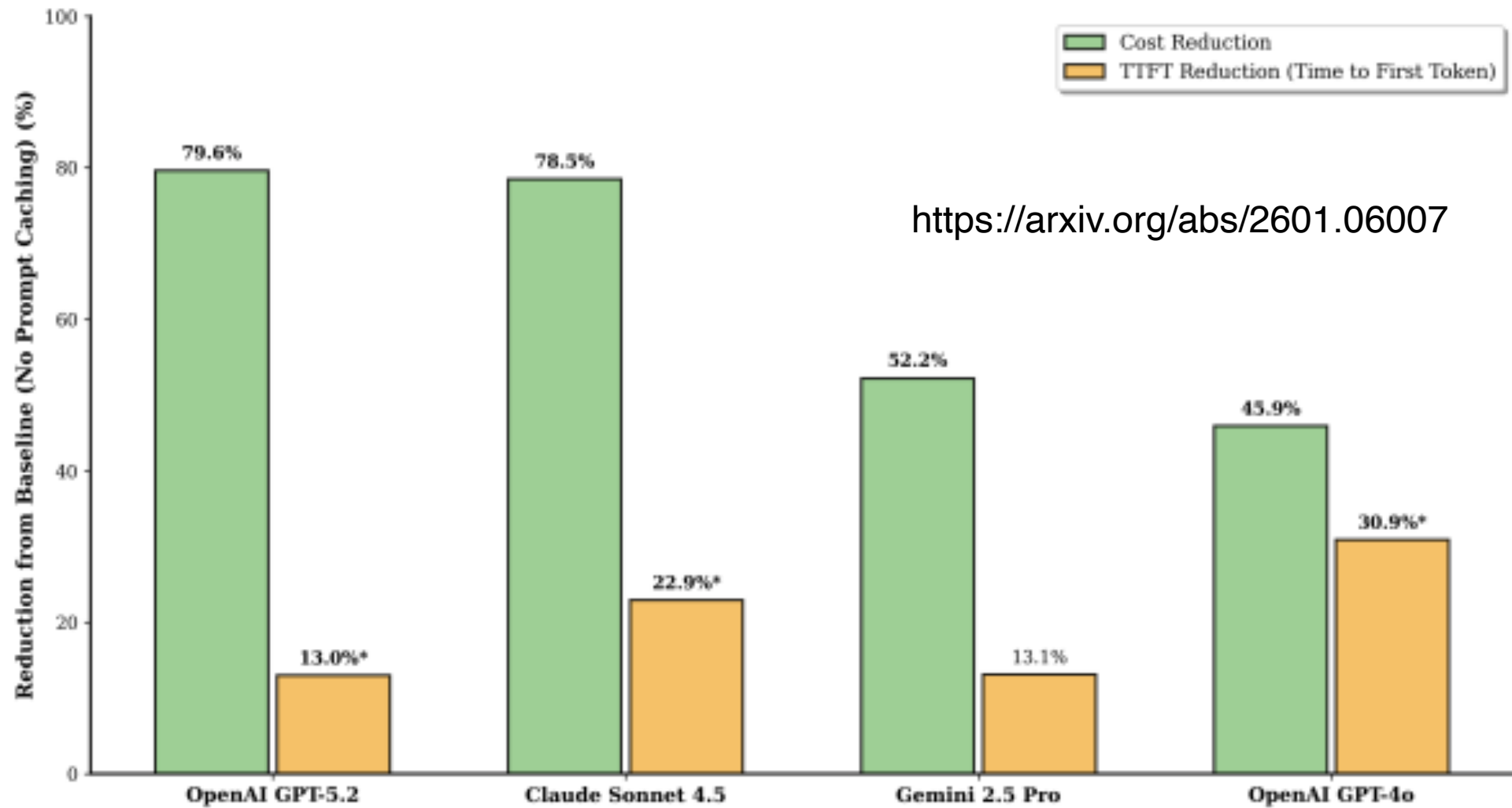
幫我訂從舊金山到紐約的班機

Cache hit more



幫我訂從 x 到 y 的班機 x =台北, y =波士頓

幫我訂從 x 到 y 的班機 x =舊金山, y =紐約



<https://arxiv.org/abs/2601.06007>

結論

	方法	是否改變原有的 Attention	是否需要訓練模型	其他代價
Flash Attention	少搬資料	X	X	一點額外運算+一點點燒腦
KV Cache	儲存已經算出來的 key 和 value	X	X	占用記憶體
Multi-query attention	多個 query 共享 key 和 value	O	O	可能明顯傷害模型能力
Group-query attention		O	O	
Multi-head Latent Attention	壓縮 key 和 value	O	O	
Sliding Window Attention	改變 Attention 範圍	O	?	
Streaming LLM		O	?	
Pruning KV Cache	丟棄 key 和 value	O	X	可能明顯傷害模型能力
Speculative Decoding	用小模型來預言生成結果	X (理論上)	X	小模型還是需要耗費額外算力